Convergence Rates of First-Order Methods: Proofs and Numerical Experiments

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Abstract

This report studies the convergence of first-order optimization methods (gradient descent, Nesterov's accelerated gradient and stochastic gradient descent) for smooth functions. For each method we prove the classical convergence rate under convexity, strong convexity or the Polyak–Łojasiewicz condition, and we compare these theoretical bounds with numerical experiments. *[Draft: to be completed.]*

1 Introduction

[To be written: motivation, question studied, contributions, outline.]

2 Setting and assumptions

We consider the unconstrained problem

$$\min_{x \in \mathbb{R}^d} f(x),$$

where $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable, and we denote by $f^* = \inf f$ and by x^* a minimizer when one exists.

Definition 1 (L-smoothness)**.** Let $L > 0$. A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is *L-smooth* if its gradient is L-Lipschitz:

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \qquad \text{for all } x, y \in \mathbb{R}^d.$$

Definition 2 (Convexity, strong convexity)**.** A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is *convex* if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \qquad \text{for all } x, y \in \mathbb{R}^d,$$

and, for $\mu > 0$, *μ-strongly convex* if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2 \qquad \text{for all } x, y \in \mathbb{R}^d.$$

When f is L-smooth and μ-strongly convex, the ratio $\kappa = L/\mu$ is called the *condition number* of f.

The Polyak–Łojasiewicz condition, a weaker assumption than strong convexity, is introduced in Section 4.2.

3 Preliminary results

Lemma 1 (Descent lemma)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth. Then for all $x, y \in \mathbb{R}^d$,*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2 .$$

Proof. Since ∇f is L-Lipschitz, it is continuous, hence f is of class C^1. Fix $x, y \in \mathbb{R}^d$ and set $\varphi(t) = f(x + t(y - x))$ for $t \in [0, 1]$. The function φ is C^1 with $\varphi'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle$, so by the fundamental theorem of calculus,

$$f(y) - f(x) = \varphi(1) - \varphi(0) = \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle \, dt.$$

Since $\int_0^1 \langle \nabla f(x), y - x \rangle \, dt = \langle \nabla f(x), y - x \rangle$, we obtain

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle = \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \, dt.$$

By the Cauchy–Schwarz inequality, then L-smoothness,

$$\begin{aligned}
\int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \, dt &\leq \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \, \|y - x\| \, dt \\
&\leq \int_0^1 L \|t(y - x)\| \, \|y - x\| \, dt \\
&= L \|y - x\|^2 \int_0^1 t \, dt \qquad \text{(since } t \geq 0\text{)} \\
&= \frac{L}{2} \|y - x\|^2 ,
\end{aligned}$$

which gives the result. □

Remark 1. Convexity is not used in the proof: the descent lemma holds for every L-smooth function. This is what allows the analysis of gradient descent for non-convex functions satisfying the Polyak–Łojasiewicz condition (Section 4.2).

Remark 2. Since the Cauchy–Schwarz inequality bounds the absolute value of the integrand, the same argument gives the two-sided bound

$$|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2} \|y - x\|^2 \qquad \text{for all } x, y \in \mathbb{R}^d.$$

Corollary 1 (Sufficient decrease)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth. For every $x \in \mathbb{R}^d$, the gradient step $x^+ = x - \frac{1}{L} \nabla f(x)$ satisfies*

$$f(x^+) \leq f(x) - \frac{1}{2L} \|\nabla f(x)\|^2 .$$

Proof. Applying Lemma 1 with $y = x^+$, so that $x^+ - x = -\frac{1}{L} \nabla f(x)$, we get

$$\begin{aligned}
f(x^+) &\leq f(x) + \left\langle \nabla f(x), -\tfrac{1}{L} \nabla f(x) \right\rangle + \frac{L}{2} \left\| -\tfrac{1}{L} \nabla f(x) \right\|^2 \\
&= f(x) - \frac{1}{L} \|\nabla f(x)\|^2 + \frac{L}{2L^2} \|\nabla f(x)\|^2 \\
&= f(x) - \frac{1}{2L} \|\nabla f(x)\|^2 .
\end{aligned}$$

□

Remark 3. In particular, each gradient step with step size $1/L$ decreases f, strictly as long as $\nabla f(x) \neq 0$. As for Lemma 1, convexity is not needed. Iterating the inequality along the gradient descent sequence $x_{k+1} = x_k - \frac{1}{L} \nabla f(x_k)$ gives the starting point of all the convergence proofs of Section 4.

4 Gradient descent

Throughout this section, $(x_k)_{k\geq 0}$ denotes the gradient descent sequence with step size $1/L$:

$$x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k), \qquad k \geq 0.$$

4.1 The quadratic case: how tight is the sufficient decrease?

Let $A \in \mathcal{S}_d(\mathbb{R})$ be symmetric positive definite, with smallest eigenvalue $\mu > 0$ and largest eigenvalue L, and let

$$f(x) = \frac{1}{2}x^\top Ax, \qquad \nabla f(x) = Ax.$$

Then f is L-smooth and μ-strongly convex, with $x^* = 0$ and $f^* = 0$. To measure how much of the guaranteed decrease of Corollary 1 is actually achieved, we define, whenever $\nabla f(x_k) \neq 0$,

$$r_k = \frac{f(x_k) - f(x_{k+1})}{\frac{1}{2L}\left\|\nabla f(x_k)\right\|^2}.$$

Corollary 1 states exactly that $r_k \geq 1$.

Proposition 1. *For every $k \geq 0$ such that $x_k \neq 0$,*

$$r_k = 2 - \frac{\langle g_k, Ag_k\rangle}{L\left\|g_k\right\|^2} \qquad \textit{where } g_k = \nabla f(x_k) = Ax_k,$$

and consequently

$$1 \leq r_k \leq 2 - \frac{\mu}{L} = 2 - \frac{1}{\kappa}.$$

Proof. Since A is invertible, $x_k \neq 0$ implies $g_k \neq 0$, so r_k is well defined. Write $x = x_k$ and $g = g_k = Ax$, so that $x_{k+1} = x - \frac{1}{L}g$.

Exact expression. By definition of r_k and of f,

$$r_k = \frac{2L}{\|g\|^2}\Big(\tfrac{1}{2}x^\top Ax - \tfrac{1}{2}(x - \tfrac{1}{L}g)^\top A(x - \tfrac{1}{L}g)\Big) = \frac{L}{\|g\|^2}\Big(\langle x, Ax\rangle - \Big\langle x - \tfrac{1}{L}g, Ax - \tfrac{1}{L}Ag\Big\rangle\Big).$$

Expanding the second inner product and using $Ax = g$,

$$\Big\langle x - \tfrac{1}{L}g, g - \tfrac{1}{L}Ag\Big\rangle = \langle x, g\rangle - \frac{1}{L}\langle x, Ag\rangle - \frac{1}{L}\langle g, g\rangle + \frac{1}{L^2}\langle g, Ag\rangle.$$

Since A is symmetric, $\langle x, Ag\rangle = \langle Ax, g\rangle = \|g\|^2$, and $\langle x, Ax\rangle = \langle x, g\rangle$. Therefore

$$r_k = \frac{L}{\|g\|^2}\Big(\frac{2}{L}\|g\|^2 - \frac{1}{L^2}\langle g, Ag\rangle\Big) = 2 - \frac{\langle g, Ag\rangle}{L\|g\|^2}.$$

Bounds. By the spectral theorem, A is diagonalizable in an orthonormal basis $(e_1, \ldots, e_d)$ of eigenvectors, with eigenvalues $\lambda_1, \ldots, \lambda_d \in [\mu, L]$. Writing $g = \sum_{i=1}^d \langle g, e_i\rangle e_i$, we get $Ag = \sum_{i=1}^d \lambda_i \langle g, e_i\rangle e_i$, hence

$$\langle g, Ag\rangle = \sum_{i=1}^d \lambda_i \langle g, e_i\rangle^2.$$

Since $\mu \leq \lambda_i \leq L$ and $\sum_{i=1}^d \langle g, e_i\rangle^2 = \|g\|^2$ (Parseval's identity in an orthonormal basis),

$$\mu\|g\|^2 \leq \langle g, Ag\rangle \leq L\|g\|^2, \qquad \text{i.e.} \qquad \frac{\langle g, Ag\rangle}{L\|g\|^2} \in \Big[\frac{\mu}{L}, 1\Big].$$

Plugging this into the exact expression gives $1 \leq r_k \leq 2 - \frac{\mu}{L}$. □

Remark 4. The quantity $\langle g, Ag\rangle / \|g\|^2$ is the Rayleigh quotient of A at g. The lower bound $r_k = 1$ is attained when g_k is an eigenvector for L: the sufficient decrease inequality is then tight. The upper bound $2 - 1/\kappa$ is attained when g_k is an eigenvector for μ. Along gradient descent, the components of x_k along the eigenvector e_i are multiplied by $(1-\lambda_i/L)$ at each step, so the slowest direction (eigenvalue μ) dominates for large k, and one expects $r_k \to 2 - 1/\kappa$. This is what we observe numerically (Section 7).

4.2 Polyak–Łojasiewicz condition

Definition 3 (Polyak–Łojasiewicz condition)**.** Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and bounded from below, with $f^\star = \inf_{x\in\mathbb{R}^d} f(x) > -\infty$. We say that f satisfies the *Polyak–Łojasiewicz (PL) condition* with constant $\mu > 0$ if

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu\,(f(x) - f^\star) \qquad \text{for all } x \in \mathbb{R}^d.$$

The PL condition does not require convexity, nor uniqueness of the minimizer: it only says that the gradient cannot be small at a point where f is far from its infimum. When f is L-smooth and satisfies the PL condition with constant μ, we set $\kappa = L/\mu$.

Theorem 1 (Linear convergence under the PL condition)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth, bounded from below with $f^\star = \inf f > -\infty$, and satisfying the PL condition with constant $\mu > 0$. Then, for any $x_0 \in \mathbb{R}^d$, the gradient descent sequence $x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$ satisfies, for all $k \geq 0$,*

$$f(x_k) - f^\star \leq \Big(1 - \frac{1}{\kappa}\Big)^k (f(x_0) - f^\star).$$

Proof. We prove the result by induction on k. For $k = 0$ there is nothing to prove. Let $k \geq 1$ and assume the bound holds at rank $k-1$. By Corollary 1 applied at $x = x_{k-1}$,

$$f(x_k) \leq f(x_{k-1}) - \frac{1}{2L}\|\nabla f(x_{k-1})\|^2.$$

By the PL condition at x_{k-1}, $\frac{1}{2L}\|\nabla f(x_{k-1})\|^2 \geq \frac{\mu}{L}(f(x_{k-1}) - f^\star)$. Subtracting $f^\star$ from both sides, we obtain

$$f(x_k) - f^\star \leq (f(x_{k-1}) - f^\star) - \frac{\mu}{L}(f(x_{k-1}) - f^\star) = \Big(1 - \frac{1}{\kappa}\Big)(f(x_{k-1}) - f^\star).$$

Since $1 - 1/\kappa \geq 0$ (see Remark 5), the induction hypothesis gives

$$f(x_k) - f^\star \leq \Big(1 - \frac{1}{\kappa}\Big)\Big(1 - \frac{1}{\kappa}\Big)^{k-1}(f(x_0) - f^\star) = \Big(1 - \frac{1}{\kappa}\Big)^k (f(x_0) - f^\star). \qquad \square$$

Remark 5. Necessarily $\mu \leq L$, so that $0 \leq 1 - 1/\kappa < 1$. Indeed, for any $x \in \mathbb{R}^d$, Corollary 1 gives $f^\star \leq f(x^+) \leq f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$, hence $\frac{1}{2L}\|\nabla f(x)\|^2 \leq f(x) - f^\star$. Combined with the PL condition, $\frac{\mu}{L}(f(x) - f^\star) \leq f(x) - f^\star$, which yields $\mu \leq L$ as soon as $f(x) > f^\star$ for some x (otherwise f is constant and the result is trivial).

Remark 6. The proof only combines the sufficient decrease of Corollary 1 with the PL inequality: convexity is never used. In particular, Theorem 1 applies to some non-convex functions, such as $x \mapsto x^2 + 3\sin^2(x)$. Moreover, strongly convex functions satisfy the PL condition (Lemma 2), so that Theorem 1 also covers the strongly convex case (Section 4.3).

4.3 Strongly convex case

Lemma 2 (Strong convexity implies PL)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and μ-strongly convex. Then f is bounded from below, and it satisfies the PL condition with the same constant μ:*

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu\,(f(x) - f^\star) \qquad \text{for all } x \in \mathbb{R}^d.$$

Proof. Fix $x \in \mathbb{R}^d$. By strong convexity, for all $y \in \mathbb{R}^d$,

$$f(y) \geq \varphi(y) := f(x) + \langle \nabla f(x), y - x\rangle + \frac{\mu}{2}\|y - x\|^2.$$

The function φ is a strictly convex quadratic function of y, whose gradient $\nabla\varphi(y) = \nabla f(x) + \mu(y - x)$ vanishes at $y^* = x - \frac{1}{\mu}\nabla f(x)$; hence y^* is its global minimizer, and

$$\min_{y\in\mathbb{R}^d} \varphi(y) = \varphi(y^*) = f(x) - \frac{1}{\mu}\|\nabla f(x)\|^2 + \frac{\mu}{2}\cdot\frac{1}{\mu^2}\|\nabla f(x)\|^2 = f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2.$$

Therefore $f(y) \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$ for all y: the right-hand side does not depend on y, so f is bounded from below, and taking the infimum over y gives

$$f^\star = \inf_{y\in\mathbb{R}^d} f(y) \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2,$$

which is the PL inequality after rearranging. □

Corollary 2 (Strongly convex case)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be L-smooth and μ-strongly convex, and $\kappa = L/\mu$. Then gradient descent with step size $1/L$ satisfies, for all $k \geq 0$,*

$$f(x_k) - f^\star \leq \Big(1 - \frac{1}{\kappa}\Big)^k (f(x_0) - f^\star).$$

Proof. By Lemma 2, f is bounded from below and satisfies the PL condition with constant μ; apply Theorem 1. □

This is the bound displayed in Figure 2.
[Optional: linear convergence of the iterates, $\|x_k - x^*\|^2 \leq (1 - 1/\kappa)^k \|x_0 - x^*\|^2$.]

4.4 Convex case

Without strong convexity (or the PL condition), linear convergence is lost in general, but gradient descent still converges at a sublinear rate, provided the infimum is attained.

Theorem 2 (Sublinear convergence in the convex case)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable, convex and L-smooth, and assume that f has a minimizer $x^\star$, with $f^\star = f(x^\star)$. Then, for any $x_0 \in \mathbb{R}^d$, the gradient descent sequence $x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$ satisfies, for all $k \geq 1$,*

$$f(x_k) - f^\star \leq \frac{L\|x_0 - x^\star\|^2}{2k}.$$

Proof. Step 1: one-step inequality. Let $i \geq 1$ and write $g = \nabla f(x_{i-1})$, so that $x_i = x_{i-1} - \frac{1}{L}g$. By convexity of f, applied at the point x_{i-1} and evaluated at $x^\star$,

$$f^\star \geq f(x_{i-1}) + \langle g, x^\star - x_{i-1}\rangle, \qquad \text{i.e.} \qquad f(x_{i-1}) - f^\star \leq \langle g, x_{i-1} - x^\star\rangle.$$

Together with Corollary 1, which gives $f(x_i) - f(x_{i-1}) \leq -\frac{1}{2L}\|g\|^2$, we get

$$f(x_i) - f^\star = \big(f(x_i) - f(x_{i-1})\big) + \big(f(x_{i-1}) - f^\star\big) \leq \langle g, x_{i-1} - x^\star\rangle - \frac{1}{2L}\|g\|^2 = \Big\langle g,\ x_{i-1} - \frac{1}{2L}g - x^\star\Big\rangle.$$

Since $\frac{1}{L}g = x_{i-1} - x_i$, we have $x_{i-1} - \frac{1}{2L}g - x^\star = \frac{1}{2}(x_{i-1} - x^\star) + \frac{1}{2}(x_i - x^\star)$ and $g = L\,(x_{i-1} - x_i) = L\big((x_{i-1} - x^\star) - (x_i - x^\star)\big)$. Hence

$$f(x_i) - f^\star \le \frac{L}{2}\Big\langle (x_{i-1} - x^\star) - (x_i - x^\star),\ (x_{i-1} - x^\star) + (x_i - x^\star)\Big\rangle = \frac{L}{2}\Big(\,\|x_{i-1} - x^\star\|^2 - \|x_i - x^\star\|^2\,\Big).$$

Step 2: summation. Let $k \ge 1$. Summing the previous inequality for $i = 1, \dots, k$, the right-hand side telescopes:

$$\sum_{i=1}^{k} \big(f(x_i) - f^\star\big) \le \frac{L}{2}\Big(\,\|x_0 - x^\star\|^2 - \|x_k - x^\star\|^2\,\Big) \le \frac{L}{2}\,\|x_0 - x^\star\|^2\,.$$

By Corollary 1, the sequence $\big(f(x_i)\big)_{i \ge 0}$ is non-increasing, so $f(x_i) \ge f(x_k)$ for all $i \in \{1, \dots, k\}$. Therefore

$$k\,\big(f(x_k) - f^\star\big) \le \sum_{i=1}^{k} \big(f(x_i) - f^\star\big) \le \frac{L}{2}\,\|x_0 - x^\star\|^2\,,$$

and dividing by k gives the result. □

Remark 7. Step 1 also shows that $\|x_i - x^\star\| \le \|x_{i-1} - x^\star\|$, since its left-hand side is non-negative: the iterates never move away from any minimizer. Moreover, the bound holds for every minimizer $x^\star$, so $\|x_0 - x^\star\|$ can be replaced by the distance from x_0 to the set of minimizers. Compared with Corollary 2, the rate $O(1/k)$ is much slower than a geometric rate: reaching accuracy ε requires $O(L\,\|x_0 - x^\star\|^2/\varepsilon)$ iterations instead of $O(\kappa \log(1/\varepsilon))$.

5 Nesterov's accelerated gradient

Theorem 2 shows that gradient descent reaches accuracy ε on convex L-smooth functions after $O(L\,\|x_0 - x^\star\|^2/\varepsilon)$ iterations. Nesterov's accelerated method improves this to $O\big(\sqrt{L/\varepsilon}\ \|x_0 - x^\star\|\,\big)$ by adding an extrapolation (momentum) step. Following [5], we present it in a form with three sequences, which makes the proof by a potential (Lyapunov) function transparent.

5.1 The convex case

Algorithm. Let f be L-smooth and let $x_0 \in \mathbb{R}^d$. Set $z_0 = x_0$ and $A_0 = 0$. For $k \ge 0$:

1. let $a_{k+1} > 0$ be the positive root of $La^2 - a - A_k = 0$, so that $L\,a_{k+1}^2 = A_k + a_{k+1}$, and set $A_{k+1} = A_k + a_{k+1}$;
2. $y_k = \dfrac{A_k}{A_{k+1}}\,x_k + \dfrac{a_{k+1}}{A_{k+1}}\,z_k$;
3. $x_{k+1} = y_k - \dfrac{1}{L}\nabla f(y_k)$;
4. $z_{k+1} = z_k - a_{k+1}\,\nabla f(y_k)$.

The point x_{k+1} is obtained by a gradient step with the usual step size $1/L$, but from the extrapolated point y_k; the sequence (z_k) accumulates all past gradients with increasing weights. Note that (A_k) is increasing, $A_{k+1} = L\,a_{k+1}^2$, and $A_1 = a_1 = 1/L$ (positive root of $La^2 = a$).

Theorem 3 (Accelerated rate in the convex case)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable, convex and L-smooth, with a minimizer $x^\star$ and $f^\star = f(x^\star)$. Then the sequence (x_k) defined above satisfies, for all $k \ge 1$,*

$$f(x_k) - f^\star \le \frac{2L\,\|x_0 - x^\star\|^2}{(k+1)^2}.$$

The proof relies on the potential

$$\Phi_k = A_k \left(f(x_k) - f^\star\right) + \frac{1}{2}\left\|z_k - x^\star\right\|^2, \qquad k \geq 0.$$

Lemma 3 (Decrease of the potential)**.** *Under the assumptions of Theorem 3, $\Phi_{k+1} \leq \Phi_k$ for all $k \geq 0$.*

Proof. Let $k \geq 0$ and $g = \nabla f(y_k)$. We split $\Phi_{k+1} - \Phi_k$ into two terms.

Function values. Using $A_{k+1} = A_k + a_{k+1}$,

$$\begin{aligned} A_{k+1}(f(x_{k+1}) - f^\star) - A_k(f(x_k) - f^\star) &= A_k(f(x_{k+1}) - f(x_k)) + a_{k+1}(f(x_{k+1}) - f^\star) \\ &= A_k(f(x_{k+1}) - f(y_k) + f(y_k) - f(x_k)) + a_{k+1}(f(x_{k+1}) - f(y_k) + f(y_k \end{aligned}$$

Since x_{k+1} is a gradient step from y_k, Corollary 1 gives $f(x_{k+1}) - f(y_k) \leq -\frac{1}{2L}\|g\|^2$. By convexity of f at y_k, evaluated at x_k and at $x^\star$,

$$f(y_k) - f(x_k) \leq \langle g, y_k - x_k\rangle, \qquad f(y_k) - f^\star \leq \langle g, y_k - x^\star\rangle.$$

Hence

$$A_{k+1}(f(x_{k+1}) - f^\star) - A_k(f(x_k) - f^\star) \leq -\frac{A_{k+1}}{2L}\|g\|^2 + \Big\langle g,\ A_k(y_k - x_k) + a_{k+1}(y_k - x^\star)\Big\rangle.$$

Distances. Since $z_{k+1} = z_k - a_{k+1}g$, expanding the square gives

$$\frac{1}{2}\|z_{k+1} - x^\star\|^2 - \frac{1}{2}\|z_k - x^\star\|^2 = \frac{a_{k+1}^2}{2}\|g\|^2 - a_{k+1}\langle g, z_k - x^\star\rangle.$$

Sum. Adding the two terms,

$$\Phi_{k+1} - \Phi_k \leq \frac{\|g\|^2}{2}\Big(a_{k+1}^2 - \frac{A_{k+1}}{L}\Big) + \Big\langle g,\ A_k(y_k - x_k) + a_{k+1}(y_k - z_k)\Big\rangle,$$

where the terms in $x^\star$ have cancelled. The first term vanishes because $La_{k+1}^2 = A_{k+1}$. For the second one, the definition of y_k gives $A_{k+1}y_k = A_k x_k + a_{k+1}z_k$, hence

$$A_k(y_k - x_k) + a_{k+1}(y_k - z_k) = A_{k+1}y_k - A_k x_k - a_{k+1}z_k = 0.$$

Therefore $\Phi_{k+1} - \Phi_k \leq 0$. □

Lemma 4 (Growth of the weights)**.** *For all $k \geq 1$, $A_k \geq \dfrac{(k+1)^2}{4L}$.*

Proof. For $k \geq 0$, $A_{k+1} - A_k = a_{k+1}$, that is $(\sqrt{A_{k+1}} + \sqrt{A_k})(\sqrt{A_{k+1}} - \sqrt{A_k}) = a_{k+1}$. Since (A_k) is increasing and $\sqrt{A_{k+1}} = \sqrt{L}\, a_{k+1}$,

$$\sqrt{A_{k+1}} - \sqrt{A_k} = \frac{a_{k+1}}{\sqrt{A_{k+1}} + \sqrt{A_k}} \geq \frac{a_{k+1}}{2\sqrt{A_{k+1}}} = \frac{1}{2\sqrt{L}}.$$

The first increment is computed exactly: $\sqrt{A_1} - \sqrt{A_0} = 1/\sqrt{L}$. Hence, for $k \geq 1$, by telescoping,

$$\sqrt{A_k} = (\sqrt{A_1} - \sqrt{A_0}) + \sum_{j=1}^{k-1}(\sqrt{A_{j+1}} - \sqrt{A_j}) \geq \frac{1}{\sqrt{L}} + \frac{k-1}{2\sqrt{L}} = \frac{k+1}{2\sqrt{L}},$$

and squaring gives the result. □

Proof of Theorem 3. By Lemma 3, (Φ_k) is non-increasing, and $\Phi_0 = \frac{1}{2}\|x_0 - x^\star\|^2$ since $A_0 = 0$ and $z_0 = x_0$. As the second term of Φ_k is non-negative, for all $k \geq 0$,

$$A_k(f(x_k) - f^\star) \leq \Phi_k \leq \Phi_0 = \frac{1}{2}\|x_0 - x^\star\|^2 .$$

For $k \geq 1$, $A_k > 0$; dividing and using Lemma 4,

$$f(x_k) - f^\star \leq \frac{\|x_0 - x^\star\|^2}{2A_k} \leq \frac{2L\,\|x_0 - x^\star\|^2}{(k+1)^2}. \qquad \square$$

Remark 8. The proof uses exactly the same ingredients as Theorem 2, namely the sufficient decrease and the convexity inequality, but combines them through a potential in which the weight A_k grows quadratically. The choice $La_{k+1}^2 = A_{k+1}$ is precisely what cancels the $\|g\|^2$ terms, and the definition of y_k is precisely what cancels the inner products. Note also that the lemma does not imply that $\|z_k - x^\star\|$ decreases, only that it stays bounded by $\|x_0 - x^\star\|$. Eliminating z_k, the method can be rewritten as a gradient step from x_k plus a momentum term proportional to $x_k - x_{k-1}$, which is the form used in our code [5].

5.2 The strongly convex case

[R3': constant momentum $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ and rate $(1 - 1/\sqrt{\kappa})^k$; this is the version implemented in the experiments.]

5.3 Lower bound and continuous-time interpretation

[Statement of the $O(1/k^2)$ lower bound for first-order methods (with reference); ODE $\ddot{x} + \frac{3}{t}\dot{x} + \nabla f(x) = 0$ [6].]

6 Stochastic gradient descent

[R5: constant step size and noise floor; decreasing step size and $O(1/k)$ rate.]

7 Numerical experiments

7.1 Gradient descent on quadratics

Setting. We consider the quadratic $f(x) = \frac{1}{2}x^\top Ax$ of Section 4.1 in dimension $d = 100$, with $\mu = 1$ and $L = \kappa$ for $\kappa \in \{10, 100, 1000\}$. The matrix A is built as $A = QDQ^\top$, where D is diagonal with entries μ, L and $d-2$ values drawn uniformly in $[\mu, L]$, and Q is the orthogonal factor of the QR factorization of a $d \times d$ matrix with i.i.d. standard Gaussian entries. By construction, the smallest and largest eigenvalues of A are exactly μ and L. The starting point x_0 is a standard Gaussian vector, and we run 5000 iterations of gradient descent with step size $1/L$. Since $f^\star = 0$, the optimality gap is $f(x_k)$ itself. Iterations with $f(x_k) < 10^{-12}$ are discarded from the plots and the statistics below: at that level, the differences $f(x_k) - f(x_{k+1})$ are dominated by floating-point rounding errors. All random quantities are drawn from a single generator with a fixed seed, and the figures are produced by the script `code/experiments/exp01_gd_quadratic.py`. Before the experiment, the gradient $\nabla f(x) = Ax$ was checked against central finite differences (maximal relative error of order 10^{-10}).

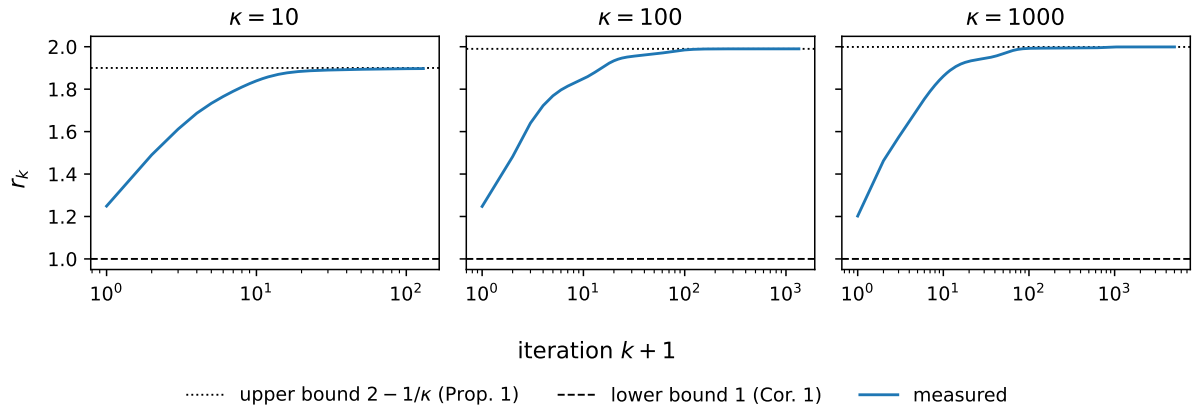


Figure 1: Ratio r_k between the actual decrease $f(x_k) - f(x_{k+1})$ and the decrease $\frac{1}{2L}\|\nabla f(x_k)\|^2$ guaranteed by Corollary 1, for $\kappa \in \{10, 100, 1000\}$ (logarithmic horizontal axis). Dashed line: lower bound 1 (Corollary 1). Dotted line: upper bound $2 - 1/\kappa$ (Proposition 1).

Tightness of the sufficient decrease (Figure 1). For the three values of κ, the ratio r_k stays in the interval $[1, 2 - 1/\kappa]$ predicted by Proposition 1. It starts at $r_0 \approx 1.25$ for $\kappa = 10$ and $\kappa = 100$, and $r_0 \approx 1.20$ for $\kappa = 1000$: at the first iterations, the guaranteed decrease of Corollary 1 is close to the actual one. Then r_k increases towards its upper bound: the maximal observed values are 1.8971, 1.9900 and 1.9990, for upper bounds 1.9, 1.99 and 1.999. For $\kappa = 10$, the remaining gap of order 3×10^{-3} is due to the eigenvalue $\lambda_2 \approx 1.03$, very close to μ, whose component has not yet vanished when the run is stopped by the threshold 10^{-12} (after 130 iterations). This is the behaviour anticipated in the remark following Proposition 1: the components of x_k along eigenvectors with large eigenvalues are damped quickly, so that $g_k = Ax_k$ aligns with the eigenvectors of the smallest eigenvalues, and the Rayleigh quotient $\langle g_k, Ag_k\rangle / \|g_k\|^2$ decreases towards μ. For $\kappa = 1000$, the convergence to $2 - 1/\kappa$ goes through an intermediate regime: the next smallest eigenvalue is $\lambda_2 \approx 6.73$, and the component along its eigenvector only decays by a factor $(1 - \lambda_2/L)/(1 - \mu/L)$ per step relative to the slowest one, that is about 0.57%. Accordingly, $r_k \approx 1.992$ at $k = 100$, close to $2 - \lambda_2/L \approx 1.993$, and r_k only reaches 1.9988 around $k = 1000$.

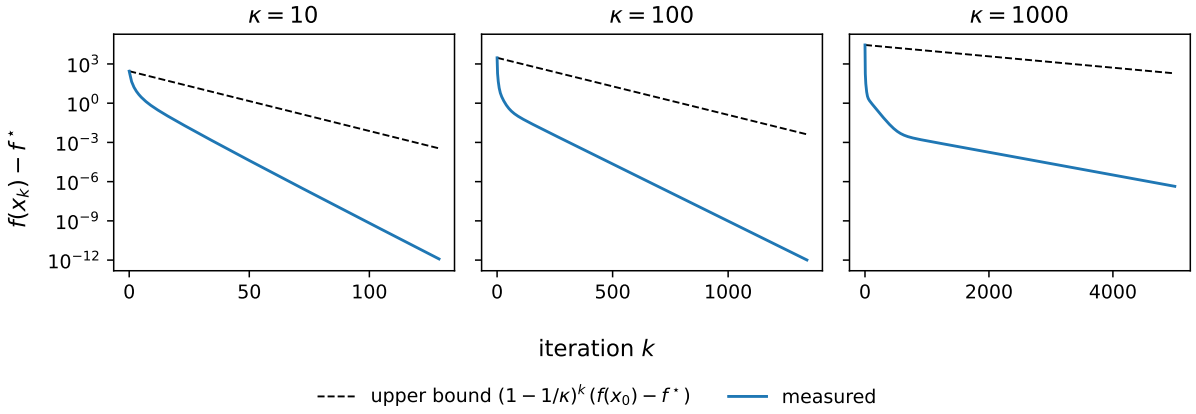


Figure 2: Optimality gap $f(x_k) - f^\star$ of gradient descent with step size $1/L$ (logarithmic vertical axis), for $\kappa \in \{10, 100, 1000\}$. Dashed line: upper bound $(1 - 1/\kappa)^k(f(x_0) - f^\star)$ of Corollary 2.

Convergence rate (Figure 2). The measured gap always stays below the bound $(1 - 1/\kappa)^k (f(x_0) - f^\star)$ of Corollary 2 (a quadratic with A positive definite is μ-strongly convex). After a short initial phase, in which the components along large eigenvalues vanish, each curve is a straight line in logarithmic scale: the convergence is linear, and its rate deteriorates as κ grows. However, the measured slope is about twice the slope of the bound. Table 1 compares the slope of $\log f(x_k)$, fitted by least squares on the second half of the plotted iterations, with the slope $\log(1 - 1/\kappa)$ of the bound. For instance, for $\kappa = 10$, the gap reaches about 10^{-12} after 130 iterations, while the bound is still of order 4×10^{-4}.

κ	measured slope	$\log(1 - 1/\kappa)$	ratio
10	-0.2186	-0.1054	2.07
100	-0.0201	-0.0101	2.00
1000	-0.0020	-0.0010	2.00

Table 1: Slope of $\log\big(f(x_k) - f^\star\big)$ as a function of k, compared with the slope of the theoretical bound.

This factor 2 has a simple heuristic explanation. Writing $x_k = \sum_i c_i^{(k)} e_i$ in an orthonormal eigenbasis of A, each component satisfies $c_i^{(k)} = (1 - \lambda_i/L)^k c_i^{(0)}$, so that

$$f(x_k) = \frac{1}{2} \sum_{i=1}^{d} \lambda_i \Big(1 - \frac{\lambda_i}{L}\Big)^{2k} \big(c_i^{(0)}\big)^2,$$

which, for large k, behaves like $C\,(1 - 1/\kappa)^{2k}$ as soon as $c_1^{(0)} \neq 0$. The exponent $2k$ comes from the fact that f is quadratic in x; the general bound, which only uses smoothness and strong convexity, loses this square. A rigorous statement is left for a forthcoming proposition.

Conclusion. On quadratics, the theory correctly predicts the qualitative behaviour of gradient descent: the sufficient decrease holds at every step, the convergence is linear and slows down as κ grows. Quantitatively, Proposition 1 describes the ratio r_k exactly, whereas the general bound on $f(x_k) - f^\star$ is pessimistic by a factor 2 in the exponent. Whether this pessimism persists in the worst case over all L-smooth and μ-strongly convex functions is a natural question, which can be investigated with performance estimation techniques.

[Further experiments: logistic regression; non-convex PL example; SGD on least squares; comparison with Adam.]

8 Discussion

[Where the bounds are tight, where they are pessimistic, and why.]

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